Real Mixed Hodge Structures

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The real mixed Hodge structures of §2 form a rigid abelian tensor category, and the functor "underlying real vector space" is a fiber functor. The functor Gr^W is another one. We compute what the general theory of Tannakian categories gives in this particular case.

1 Three opposite filtrations

1.1

This paragraph is a variation on §1.2 of P. Deligne, *Théorie de Hodge II*, Publ. Math. IHES 40, pp. 5–58. Let $\mathcal{A}$ be an abelian category, and let A be an object of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \overline{F}$. Here "filtration" means finite filtration; W is increasing, and F and $\overline{F}$ are decreasing. The condition of opposition is

$$\mathrm{Gr}_n^W \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q A = 0 \qquad \text{for } n \neq p+q.$$

For a bigraded object $B = \bigoplus B^{p,q}$, we write $W, F_W, \overline{F}_W$ for the three opposite filtrations

$$W_n B = \bigoplus_{p+q \leq n} B^{p,q}, \qquad F_W^i B = \bigoplus_{p \geq i} B^{p,q}, \qquad \overline{F}_W^i B = \bigoplus_{q \geq i} B^{p,q}.$$

With this notation, saying that $W, F, \overline{F}$ are opposite on A is equivalent to saying that, for each n, $\mathrm{Gr}_n^W A$ has a unique decomposition

$$\mathrm{Gr}_n^W A = \bigoplus_{p+q=n} A_W^{p,q}$$

such that the filtrations induced by F and $\overline{F}$ on $\mathrm{Gr}_n^W A$ are given by

$$F^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n,\, p \geq i} A_W^{p,q}, \qquad \overline{F}^i \mathrm{Gr}_n^W A = \bigoplus_{p+q=n,\, q \geq i} A_W^{p,q}.$$

Put

$$A_F^{p,q} := W_{p+q} A \cap F^p A \cap \left(\overline{F}^q A + \sum_{i>0} W_{p+q-i} A \cap \overline{F}^{q-i+1} A \right),$$

and, interchanging the roles of $F, \overline{F}$ and of p, q,

$$A_{\overline{F}}^{p,q} := W_{p+q} A \cap \overline{F}^q A \cap \left(F^p A + \sum_{i>0} W_{p+q-i} A \cap F^{p-i+1} A \right).$$

For $p+q=n$, the projection from $W_n A$ to $\mathrm{Gr}_n^W A$ induces isomorphisms

$$A_F^{p,q} \simeq A_W^{p,q}, \qquad A_{\overline{F}}^{p,q} \simeq A_W^{p,q}.$$

It follows that the $A_F^{p,q}$, respectively the $A_{\overline{F}}^{p,q}$, form a bigrading of A. The sums of the isomorphisms

$$A_F^{p,q} \xrightarrow{\sim} A_W^{p,q}, \qquad A_{\overline{F}}^{p,q} \xrightarrow{\sim} A_W^{p,q}$$

are bifiltered isomorphisms

$$a_F : (A; W, F) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, F_W), \qquad a_{\overline{F}} : (A; W, \overline{F}) \xrightarrow{\sim} (\mathrm{Gr}^W A; W, \overline{F}_W).$$

Lemma 1.1. *The automorphism $d = a_F a_{\overline{F}}^{-1}$ of $\operatorname{Gr}^W A$ satisfies*

$$(d-1)(A_W^{p,q}) \subset \bigoplus_{\substack{r<p\\ s<q}} A_W^{r,s}. \tag{1.1.1}$$

Proof. Passing to the associated graded, the isomorphisms a_F and $a_{\overline{F}}$ induce the identity on $\operatorname{Gr}^W A$. Thus $\operatorname{Gr}^W(d) = 1$. For a subset $I \subset \mathbb{Z}^2$ and a bigraded object B, write $B_I = \bigoplus_{(r,s)\in I} B^{r,s}$. Let

$$I(p,q) = \{(r,s) : r+s \le p+q \text{ and } ((r,s)=(p,q) \text{ or } r<p)\},$$
$$J(p,q) = \{(r,s) : r+s \le p+q \text{ and } ((r,s)=(p,q) \text{ or } s<q)\},$$
$$K(p,q) = \{(r,s) : r+s \le p+q \text{ and } r<p \text{ and } s<q\}.$$

In the definition of $A_F^{p,q}$ as an intersection of two subobjects, the first subobject is $A_{F\,I(p,q)}$ and the second is $A_{\overline{F}\,J(p,q)}$. Hence $dA_W^{p,q}$ lies in the intersection corresponding to $I(p,q) \cap J(p,q) = K(p,q) \cup \{(p,q)\}$. Since $\operatorname{Gr}^W(d) = 1$, this gives exactly (1.1.1). □

Proposition 1.1. *The functor $A \mapsto \operatorname{Gr}^W A$ is an equivalence from the category of objects of $\mathcal{A}$ endowed with three opposite filtrations $W, F, \overline{F}$ to the category of objects of $\mathcal{A}$ endowed with a finite bigrading and with an automorphism d satisfying* (1.1.1).

The isomorphism $a_F : A \to \operatorname{Gr}^W A$ respects W, sends F to F_W, and sends $\overline{F}$ to $d^{-1}\overline{F}_W$. Thus the functor Gr^W has the inverse

$$s : (B = \bigoplus B^{p,q}, d) \longmapsto (B, W, F_W, d^{-1}\overline{F}_W),$$

with the functorial isomorphism $a_F : A \xrightarrow{\sim} s \operatorname{Gr}^W A$. Conversely, starting from a bigraded object B with an automorphism d, define $W, F, \overline{F}$ as above. If d satisfies (1.1.1), the evident map $B \to \operatorname{Gr}^W B$ is an isomorphism of functors $\mathrm{Id} \xrightarrow{\sim} \operatorname{Gr}^W \circ s$.

Remark 1.1. *If* 2 *is invertible in the ground field, then d has a unique square root $d^{1/2}$ still satisfying* (1.1.1). *Put*

$$a := d^{1/2} a_F = d^{-1/2} a_{\overline{F}}.$$

Then $a_{\overline{F}} = d^{1/2} a$ and $a_F = d^{-1/2} a$; thus a sends F, respectively $\overline{F}$, to

$$d^{1/2}(F_W), \qquad d^{-1/2}(\overline{F}_W)$$

on $\operatorname{Gr}^W A$. The isomorphism a is the same for $(A; W, F, \overline{F})$ and for $(A; W, \overline{F}, F)$.

1.4

The constructions are compatible with tensor products. If

$$\otimes : \mathcal{A}_1 \times \mathcal{A}_2 \to \mathcal{A}$$

is a biadditive exact functor, it is extended to filtered objects by

$$F^n(A_1 \otimes A_2) = \sum_{a+b=n} F^a A_1 \otimes F^b A_2 \subset A_1 \otimes A_2.$$

The associated-graded functor is compatible with $\otimes$, and, for bifiltered objects, the canonical isomorphism

$$\operatorname{Gr}^W \operatorname{Gr}^F (A_1 \otimes A_2) \simeq \operatorname{Gr}^W \operatorname{Gr}^F (A_1) \otimes \operatorname{Gr}^W \operatorname{Gr}^F (A_2)$$

sends the filtration induced by $F_1 \otimes F_2$ to the tensor product of the filtrations induced by F_1 and F_2. For vector spaces this follows from the fact that every bifiltration underlies a bigrading. Hence, if $(W_i, F_i, \overline{F}_i)$ are three opposite filtrations on A_i, then $(W_1 \otimes W_2, F_1 \otimes F_2, \overline{F}_1 \otimes \overline{F}_2)$ are again opposite on $A_1 \otimes A_2$. The bigradings, the isomorphisms $a_F, a_{\overline{F}}$, and the automorphism d are all compatible with $\otimes$.

1.5

Let k be a commutative field, let $\mathcal{A}$ be the category of finite-dimensional vector spaces over k, and let $\mathcal{C}$ be the category of vector spaces endowed with three opposite filtrations $W, F, \overline{F}$. This is a rigid abelian tensor category with $\mathrm{End}(1) = k$. It has the fiber functor ω_0, the underlying vector space, hence it is neutral Tannakian. It also has the fiber functor $\mathrm{Gr}^W =: \omega_W$. The isomorphisms a_F and $a_{\overline{F}}$ are isomorphisms of fiber functors $\omega_0 \to \omega_W$. Put

$$\mathcal{G} = \mathrm{Aut}(\omega_W).$$

The functorial automorphism d defines a k-point of the pro-algebraic group $\mathcal{G}$. The functorial bigrading of $\mathrm{Gr}^W(V)$ by the $V^{p,q}$ corresponds to a homomorphism

$$\alpha : \mathbb{G}_m \times \mathbb{G}_m \longrightarrow \mathcal{G}.$$

It is a section of the projection

$$\mathrm{pr} : \mathcal{G} \longrightarrow \mathbb{G}_m \times \mathbb{G}_m$$

corresponding to the inclusion of the category of bigraded vector spaces in $\mathcal{C}$. We normalize α by

$$\alpha(\lambda, \mu)v = \lambda^p \mu^q v \qquad (v \in V^{p,q}).$$

By Proposition 1.1, $\mathcal{G}$ is also the algebraic group of automorphisms of the fiber functor on the category of bigraded vector spaces endowed with an automorphism d satisfying (1.1.1). We now compute it when k has characteristic zero.

Let $\mathfrak{l}$ be the Lie algebra over k freely generated by elements

$$D^{i,j} \qquad (i, j < 0),$$

with the unique bigrading in which $D^{i,j}$ has bidegree (i, j). The $W_n\mathfrak{l}$ are ideals, and the quotients $\mathfrak{l}/W_n\mathfrak{l}$ are nilpotent Lie algebras. Let $\mathcal{U}$ be the pro-algebraic group which is the projective limit of the unipotent groups

$$\exp(\mathfrak{l}/W_n\mathfrak{l}).$$

Let $\mathbb{G}_m \times \mathbb{G}_m$ act on $\mathfrak{l}$ so that (λ, μ) acts on bidegree (i, j) by multiplication by $\lambda^i \mu^j$. This action respects W and therefore gives an action of $\mathbb{G}_m \times \mathbb{G}_m$ on $\mathcal{U}$.

Construction 1.1. *The group $\mathcal{G}$ is the semidirect product*

$$\mathcal{G} = (\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U}.$$

Indeed, giving an action of this semidirect product on a finite-dimensional vector space V is equivalent to giving an action of $\mathbb{G}_m \times \mathbb{G}_m$, hence a bigrading, together with a compatible nilpotent action of $\mathfrak{l}$. Equivalently, it is the datum of endomorphisms $D^{i,j}$ of bidegree (i, j), or of their sum

$$D = \sum D^{i,j},$$

with

$$D(V^{p,q}) \subset \bigoplus_{r<p,\, s<q} V^{r,s}.$$

Writing $d = \exp(D)$, this is precisely the datum of a bigrading and of an automorphism satisfying (1.1.1). *The construction is compatible with tensor products, giving the required isomorphism*

$$(\mathbb{G}_m \times \mathbb{G}_m) \ltimes \mathcal{U} \xrightarrow{\sim} \mathcal{G} = \mathrm{Aut}(\omega_W).$$

2 Mixed Hodge structures

Recall that a real mixed Hodge structure is a finite-dimensional real vector space endowed with an increasing filtration W, whose complexification $V_{\mathbb{C}}$ is endowed with a decreasing filtration F, such that the filtration $W_{\mathbb{C}}$, the filtration F, and its complex conjugate $\overline{F}$ form a system of three opposite filtrations.

Via the dictionary between real vector spaces and complex vector spaces with a real structure, giving a real mixed Hodge structure is equivalently giving a complex vector space with three opposite filtrations $W, F, \overline{F}$ and with a complex conjugation respecting W and interchanging F and $\overline{F}$. By Proposition 1.1, the functor Gr^W identifies this category with the category of bigraded complex vector spaces

$$V = \bigoplus V^{p,q}$$

equipped with an automorphism d satisfying (1.1.1) and a complex conjugation σ interchanging $V^{p,q}$ and $V^{q,p}$, for which the complex conjugate of d is d^{-1}:

$$\sigma d \sigma^{-1} = d^{-1}.$$

Let $\mathcal{H}$ be the category of real mixed Hodge structures. It is a rigid abelian tensor category with $\mathrm{End}(1) = \mathbb{R}$. It has the fiber functor ω_0, the underlying real vector space, hence is neutral Tannakian. It also has the fiber functor $\mathrm{Gr}^W =: \omega_W$. With the notation of Remark 1.1, the functorial morphism

$$a : V_{\mathbb{C}} \longrightarrow \mathrm{Gr}^W(V_{\mathbb{C}})$$

respects the real structures on both sides, since it depends symmetrically on F and $\overline{F}$. It induces an isomorphism of fiber functors, again denoted

$$a : \omega_0 \xrightarrow{\sim} \omega_W.$$

With the notation of Section 1, put $k = \mathbb{C}$, and let $\mathcal{G}_{\mathbb{R}}$ be the real form of $\mathcal{G}$ defined by the following complex conjugation:

(a) on $\mathbb{G}_m \times \mathbb{G}_m$, it is $(\lambda, \mu) \mapsto (\overline{\mu}, \overline{\lambda})$;

(b) on $\mathcal{U}$, it is the one compatible with the complex conjugation of $\mathfrak{l}$ given by

$$D^{i,j} \longmapsto -D^{j,i}.$$

As in §1.1, giving an action of $\mathcal{G}_{\mathbb{R}}$ on a real vector space V is equivalent to giving:

(a) a bigrading $V_{\mathbb{C}} = \bigoplus V^{p,q}$ such that $V^{p,q}$ and $V^{q,p}$ are complex conjugates of one another;

(b) an automorphism d of $V_{\mathbb{C}}$ such that $\overline{d} = d^{-1}$ and

$$(d-1)V^{p,q} \subset \bigoplus_{i<p,\, j<q} V^{i,j}.$$

Proposition 2.1. *The fiber functor ω_W induces an equivalence between $\mathcal{H}$ and the category of representations of $\mathcal{G}_{\mathbb{R}}$. Thus*

$$\mathcal{G}_{\mathbb{R}} = \mathrm{Aut}(\omega_W).$$

The inverse functor is

$$(V,\ \textit{action of } \mathcal{G}_{\mathbb{R}}) \longmapsto (V,\ \textit{bigrading of } V_{\mathbb{C}}, d) \longmapsto (V, W, F),$$

where W is the real filtration whose complexification is the weight filtration attached to the bigrading of $V_{\mathbb{C}}$, and

$$F = d^{-1/2}(F_W)$$

on the complexification. The isomorphism a gives the functorial identification

$$\phi \circ \mathrm{Gr}^W \xrightarrow{\sim} \mathrm{Id}.$$

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